

Basic Level Analysis

Question 1

What is the overall employee attrition rate?

Analysis

The dataset consists of **1,470 employees**, out of which **237 employees** have left the organization. This results in an overall employee attrition rate of **16.12%**.

Insight

The organization experiences a moderate level of employee turnover, with approximately one out of every six employees leaving the company.

Recommendation

Regularly monitor attrition trends and identify departments or employee groups with higher turnover to implement targeted retention strategies.

Question 2

Which department has the highest employee attrition?

Analysis

The **Sales** department recorded the highest attrition rate at **20.63%**, followed by **Human Resources (19.05%)** and **Research & Development (13.84%)**. This indicates that employee turnover varies significantly across departments.

Insight

Employees in the Sales department are leaving the organization at a higher rate than those in other departments, suggesting department-specific challenges.

Recommendation

Review employee workload, career growth opportunities, and engagement initiatives within the Sales department to improve employee retention.

Question 3

What is the average age of employees who have left the company?

Analysis

The average age of employees who left the organization is **33.61 years**, indicating that attrition is more common among employees in the early to mid stages of their careers.

Insight

The organization is experiencing higher turnover among relatively younger employees, who may be seeking better career opportunities or professional growth.

Recommendation

Develop career progression plans, mentoring programs, and employee development initiatives to improve retention among younger employees.

Question 4

Which job role has the highest average job satisfaction?

Analysis

Healthcare Representatives reported the highest average job satisfaction score of **2.79**, followed closely by **Research Scientists (2.77)** and **Sales Executives (2.75)**. The overall average job satisfaction across the organization is **2.73**.

Insight

Job satisfaction levels are relatively consistent across job roles, with only minor differences observed between positions.

Recommendation

Continue monitoring employee satisfaction through regular surveys and strengthen engagement initiatives across all job roles to maintain a positive work environment.

Question 5

Is there a significant difference in attrition rates between male and female employees?

Analysis

The attrition rate among **male employees** is **17.01%**, while **female employees** have an attrition rate of **14.80%**. Although male employees show a slightly higher turnover, the difference between the two groups is relatively small.

Insight

Gender does not appear to be a major factor influencing employee attrition, suggesting that other organizational or job-related factors play a more significant role.

Recommendation

Focus retention initiatives on employee engagement, career development, and workplace satisfaction rather than designing gender-specific retention strategies.

Question 6

What is the average monthly income of employees who have left the company?

Analysis

The average monthly income of employees who left the organization is **₹4,787.09**. This indicates that employees with comparatively lower salaries are more likely to leave the company.

Insight

Lower income levels may contribute to employee dissatisfaction and increase the likelihood of seeking better-paying opportunities.

Recommendation

Review the compensation structure and ensure salaries remain competitive with industry standards while providing performance-based rewards and career growth opportunities.

Question 7

How does distance from home impact employee attrition?

Analysis

Employee attrition rates vary across different commuting distances. Employees living **24 km from the workplace** recorded the highest attrition rate of **42.86%**, while several other distance groups also showed above-average attrition. However, no consistent increasing or decreasing trend was observed across all distance categories.

Insight

Commuting distance may influence employee turnover for certain groups, but it is not the sole factor driving attrition.

Recommendation

Consider flexible work arrangements, hybrid work options, or transportation support for employees with longer commuting distances to improve retention.

Question 8

What is the distribution of performance ratings among employees?

Analysis

Out of **1,470 employees**, **1,244 employees** received a **Performance Rating of 3**, while **226 employees** received a **Performance Rating of 4**. The majority of employees fall within the standard performance category.

Insight

Employee performance ratings are concentrated around Rating 3, indicating relatively consistent performance across the organization with a smaller group of high-performing employees.

Recommendation

Introduce recognition and development programs to encourage more employees to achieve higher performance ratings while maintaining fairness in performance evaluations.

Question 9

How many employees work overtime regularly?

Analysis

A total of **416 employees** regularly work overtime, while **1,054 employees** do not. Approximately **28%** of the workforce works overtime on a regular basis.

Insight

A significant portion of employees work beyond their regular hours, which may affect work-life balance and employee well-being if sustained over long periods.

Recommendation

Monitor overtime patterns regularly and ensure workloads are distributed effectively to prevent employee burnout and improve work-life balance.

Question 10

What is the average number of years employees have worked at the company?

Analysis

Employees have worked at the organization for an average of **7.01 years**, indicating a moderately experienced workforce with a balanced mix of newer and long-serving employees.

Insight

The average tenure suggests reasonable employee stability; however, retention efforts should continue to focus on employees during their early years of employment.

Recommendation

Strengthen onboarding, career development, and employee engagement programs to encourage long-term retention and support continuous professional growth.

Medium Level Analysis

Question 11

What are the top three factors contributing to employee attrition?

Analysis

Correlation analysis identified **Monthly Income (-0.160)**, **Age (-0.159)**, and **Years at Company (-0.134)** as the top three factors associated with employee attrition. All three variables show a weak negative correlation, indicating that employees with lower income, younger age, and shorter tenure are slightly more likely to leave the organization.

Insight

Although the correlations are not strong, compensation, employee experience, and tenure appear to have the greatest influence on employee attrition among the factors analyzed.

Recommendation

Strengthen compensation policies, career development opportunities, and employee retention programs, especially for newer and younger employees.

Question 12

How does the attrition rate vary with different levels of job involvement?

Analysis

Employees with **Job Involvement Level 1** recorded the highest attrition rate (**33.73%**), while attrition steadily decreased as involvement increased, reaching only **9.03%** at Level 4.

Insight

Higher job involvement is strongly associated with lower employee attrition, indicating that engaged employees are more likely to remain with the organization.

Recommendation

Increase employee engagement through recognition programs, meaningful work assignments, and regular performance discussions.

Question 13

Is there a relationship between employee age and job satisfaction?

Analysis

The correlation between employee age and job satisfaction is **-0.005**, indicating virtually no relationship between these two variables.

Insight

Employee satisfaction appears to be independent of age, suggesting that employees across different age groups experience similar satisfaction levels.

Recommendation

Focus on workplace culture, career development, and employee engagement initiatives rather than age-specific satisfaction programs.

Question 14

How does marital status impact employee attrition?

Analysis

Single employees recorded the highest attrition rate at **25.53%**, compared to **Married employees (12.48%)** and **Divorced employees (10.09%)**.

Insight

Single employees are considerably more likely to leave the organization, making them a higher-risk employee group.

Recommendation

Provide career growth opportunities, mentoring, and employee engagement initiatives to improve retention among single employees.

Question 15

What is the impact of training on employee attrition?

Analysis

Employees who received **no training** experienced the highest attrition rate (**27.78%**). In general, employees who participated in more training programs showed lower attrition rates, although minor variations were observed across some training levels.

Insight

Employee training and professional development contribute positively to employee retention by improving engagement and career growth opportunities.

Recommendation

Expand learning and development programs and encourage employees to participate in regular training sessions.

Question 16

How does work-life balance affect employee performance ratings?

Analysis

The percentage of employees receiving a **Performance Rating of 4** remains relatively consistent across all work-life balance levels, ranging from **13.95% to 17.50%**.

Insight

Work-life balance does not appear to have a significant impact on employee performance ratings within this dataset.

Recommendation

Continue supporting work-life balance initiatives while recognizing that employee performance is likely influenced by multiple organizational factors.

Question 17

What is the effect of stock options on employee attrition?

Analysis

Employees with **no stock options (Level 0)** recorded the highest attrition rate (**24.41%**). Attrition decreased significantly for employees with **Stock Option Levels 1 and 2**, before increasing slightly at Level 3.

Insight

Providing stock options appears to improve employee retention and strengthen long-term commitment to the organization.

Recommendation

Review employee compensation packages and consider expanding stock-based incentives to improve retention.

Question 18

How does employee tenure relate to job satisfaction?

Analysis

The correlation between employee tenure and job satisfaction is approximately - **0.004**, indicating no meaningful relationship between the two variables.

Insight

Employees with longer service do not necessarily report higher or lower job satisfaction than newer employees.

Recommendation

Maintain consistent employee engagement and satisfaction initiatives throughout the employee lifecycle rather than focusing solely on tenure.

Question 19

What is the impact of business travel on job satisfaction?

Analysis

Average job satisfaction scores are very similar across all travel categories, with **Non-Travel** and **Travel Frequently** employees averaging **2.79**, while **Travel Rarely** employees average **2.70**.

Insight

Business travel frequency has only a minimal influence on employee job satisfaction.

Recommendation

Continue offering balanced travel policies while prioritizing broader workplace improvements that have a greater impact on employee satisfaction.

Question 20

How do years since the last promotion affect employee performance?

Analysis

Employees promoted within the last year recorded **16.52%** of high performance ratings, while other promotion intervals showed only minor variations. Categories with longer promotion gaps contained relatively few employees, making those percentages less reliable.

Insight

No consistent relationship was observed between years since the last promotion and employee performance.

Recommendation

Continue promoting employees based on merit and performance while supporting career growth through regular feedback, recognition, and development opportunities.

Overall Medium Analysis Summary

The medium-level analysis indicates that **job involvement, stock options, training opportunities, and marital status** have the strongest association with employee attrition. In contrast, factors such as **age, tenure, business travel, work-life balance, and years since the last promotion** show little or no meaningful relationship with employee satisfaction or performance. These findings suggest that organizations should prioritize employee engagement, career development, and competitive compensation strategies to improve long-term retention.